

Technical Document #346: Natural Language Processing - Comprehensive Overview

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Text summarization generates a concise summary of a longer text. Extractive summarization selects important sentences while abstractive summarization generates new text.

Machine translation converts text from one language to another. Neural machine translation using encoder-decoder architectures has achieved human-level performance.

Question answering systems extract answers from a given context. They typically consist of a retriever component and a reader component.

Coreference resolution identifies all expressions that refer to the same entity in a text. It is essential for understanding document-level coherence.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

The rapid advancement of technology has transformed how organizations approach problem-solving and innovation. Companies must adapt to stay competitive in an ever-evolving landscape.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Key Takeaways:

- Tokenization is the process of breaking text into individual units called tokens.
- Coreference resolution identifies all expressions that refer to the same entity in a text.
- Language models predict the probability of a sequence of words.